P0 — Project Bootstrap Framework

Extension for the A2C Architecture-to-Code Framework

Phase Zero scaffolding: project structure, build manifest, git config, env templates, Docker, Makefile

Built on: A2C Framework | E2A Architecture Framework

github.com/subhamviky/a2c-framework

Author: Subham Gupta, Staff Architect & AI Architect

1. EXECUTIVE SUMMARY

The A2C Framework generates enterprise-grade microservice code, Terraform IaC, and GitHub Actions CI/CD pipelines using E2A-governed agents. However, before A2C's RequirementsAgent can begin, the target project must exist as a valid, compilable workspace: a Maven project with a pom.xml and correct directory tree, or a Poetry project with pyproject.toml and package structure. This Phase Zero bootstrapping was previously manual — a gap between specification and generation.

The P0 Framework fills this gap. It is a new abstract class alongside A2C — not an extension of it — that provides a single public entry point to generate everything a project needs before any AI-based code generation begins. The developer provides a ScaffoldRequest (from a JSON config file, a natural-language prompt, or an inline dict). P0 creates the entire project scaffold: directory tree, build manifest, dependency configuration, git rules, environment templates, README skeleton, LICENSE, Dockerfile, and Makefile.

Architecture Decision — New Class, Not Extension

P0 and A2C are separate concerns on different lifecycles. Bootstrap runs once per project; A2C runs whenever a new service or component is generated. Making P0 an extension of A2C would force bootstrap to execute on every A2C call. The correct design: two independent classes composable via BootstrapAndGenerateWorkflow, each callable alone or in sequence.

Aspect	P0 Framework	A2C Framework
Purpose	Generate project structure, build manifest, git config, Docker, Makefile	Generate microservice code, Terraform IaC, GitHub Actions CI/CD
Input	ScaffoldRequest (runtime, build_tool, platform, dependencies)	DevRequest (project_type, mandatory_nfrs, target_cloud, context)
Public entry point	bootstrapper.bootstrap(request, config)	sdlc_agent.run(dev_request, config)
Output	ScaffoldResult (scaffolded_files, project_root, build_manifest_path)	DevRequest (generated_code, generated_iac, generated_cicd, critic_report)
Runs when	Once per project — at project creation	As needed — whenever a new component is generated
Lifecycle	Phase Zero — before any business logic	Phase 2-7 — NFR-First development sequence
NFR enforcement	Structural: correct gitignore, non-root Docker, HEALTHCHECK, Makefile targets	Contract: _apply_policy() injects idempotency, circuit breakers, observability
Composable?	Yes — BootstrapAndGenerateWorkflow chains P0 -> A2C	Yes — chains from P0 result via dev_request handoff

2. FRAMEWORK ARCHITECTURE

2.1 Class Hierarchy

P0 follows the same E2A Template Method Pattern as A2C and E2A. One PUBLIC method (bootstrap) contains the full orchestration sequence. All domain-specific behaviour is in PROTECTED methods that subclasses override. Infrastructure mechanics (file I/O, idempotency, config loading) are in PRIVATE methods that subclasses never touch.

```
BaseProjectBootstrapper (ABC)          — abstract base, one public entry point
  |
  +-- PythonPoetryBootstrapper         — Python 3.x + Poetry
  +-- PythonPipBootstrapper            — Python 3.x + pip / setup.cfg
  +-- JavaMavenBootstrapper            — Java 21 + Maven / Spring Boot
  +-- JavaGradleBootstrapper           — Java 21 + Gradle (Kotlin DSL)
  +-- NodeNpmBootstrapper              — Node.js + npm / package.json
  +-- GoModulesBootstrapper            — Go + go.mod

ProjectBootstrapperFactory             — resolves subclass from ScaffoldRequest

BootstrapAndGenerateWorkflow           — orchestrates P0 -> A2C in sequence
```

2.2 TypedDicts

ScaffoldRequest — input to bootstrap()

Field	Type	Required	Description
source	str	No	'json' | 'prompt' | 'inline' (default: inline)
config_path	str	If source='json'	Path to scaffold-config.json
prompt	str	If source='prompt'	Free-text description for LLM parsing
runtime	str	Yes	'python' | 'java' | 'node' | 'go'
runtime_version	str	No	'3.12' | '21' | '20' | '1.22'
platform	str	Yes	'aws' | 'gcp' | 'azure' | 'standalone'
project_name	str	Yes	PascalCase: 'SettlementEngine'
package_name	str	Python only	snake_case: 'settlement_engine'
group_id	str	Java only	'com.company.domain'
artifact_id	str	Java only	kebab-case: 'settlement-engine'
build_tool	str	Yes	'poetry' | 'pip' | 'maven' | 'gradle' | 'npm' | 'go_modules'
project_type	str	No	'fastapi_microservice' | 'spring_boot' | 'lambda' | 'langgraph_agent' | 'library'
dependencies	List[str]	No	Runtime deps: ['fastapi>=0.111', 'pydantic>=2.7']
dev_dependencies	List[str]	No	Dev/test deps: ['pytest>=8', 'ruff>=0.4']
spring_boot_version	str	Java only	Default: '3.3.0'
java_source_version	str	Java only	Default: '21'
output_path	str	No	Base output dir (default: './')
license	str	No	'Apache-2.0' | 'MIT' | 'PROPRIETARY' (default: Apache-2.0)
include_docker	bool	No	Generate Dockerfile + .dockerignore (default: true)
include_makefile	bool	No	Generate Makefile (default: true)

ScaffoldResult — output from bootstrap()

Field	Type	Description
success	bool	True if all required files were created and validated
scaffold_key	str	Business idempotency key: name:runtime:build_tool:platform
project_root	str	Absolute path to created project root directory
scaffolded_files	List[str]	Absolute paths of every file created
build_manifest_path	str	Path to pom.xml / pyproject.toml / go.mod
readme_path	str	Path to generated README.md
trace_id	str	UUID correlation ID for observability
latency_ms	float	Total execution time in milliseconds
errors	List[str]	Error messages if success=False
dev_request	dict | None	A2C DevRequest — populated when config['a2c']['enabled']=True

3. BASEPROJECTBOOTSTRAPPER — FULL CLASS CONTRACT

3.1 Class Variables

Variable	Type	Default	Purpose
bootstrapper_name	str	'BaseProjectBootstrapper'	Governance key — set in every subclass. Used in all structured log entries.
idempotency	bool	True	When True: skip if project directory already exists and is non-empty. Set False to re-scaffold.
APPROVED_RUNTIMES	List[str]	['python','java','node','go']	Governance allow-list for runtime field.
APPROVED_PLATFORMS	List[str]	['aws','gcp','azure','standalone']	Governance allow-list for platform field.
APPROVED_BUILD_TOOLS	Dict[str,List[str]]	See class	Per-runtime allow-list for build_tool field.
MANIFEST_FILENAMES	Dict[tuple,str]	See class	Maps (runtime, build_tool) to correct manifest filename.

3.2 Method Contract Table

ACCESS	METHOD SIGNATURE	RETURNS	PURPOSE / WHEN TO OVERRIDE
PUBLIC	`bootstrap(request, config, **kwargs)`	`ScaffoldResult`	Single entry point. Never override. Full 16-step sequence.
ABSTRACT	`_validate_request(request, config, **kwargs)`	`bool`	Check required fields, approved runtime/platform/build_tool. Return False to abort.
ABSTRACT	`_resolve_config(request, config, **kwargs)`	`ScaffoldRequest`	Load from JSON file or LLM prompt. Merge into request dict.
ABSTRACT	`_plan_scaffold(request, config, **kwargs)`	`List[dict]`	Build [{step, file, description}] task list for observability.
ABSTRACT	`_scaffold_structure(request, config, root, **kwargs)`	`None`	Create all directories. Call __create_directory() per path.
ABSTRACT	`_generate_build_manifest(request, config, **kwargs)`	`str`	Return pom.xml | pyproject.toml | go.mod content as string.
PROTECTED	`_generate_dependency_config(request, config, **kwargs)`	`str|None`	Return requirements.txt content or None. Default: None.
PROTECTED	`_dependency_file_name(request)`	`str`	Filename for dep config. Default: 'requirements.txt'.
ABSTRACT	`_generate_git_config(request, config, **kwargs)`	`Dict[str,str]`	Return {'.gitignore': content, '.gitattributes': content}.
ABSTRACT	`_generate_env_templates(request, config, **kwargs)`	`Dict[str,str]`	Return {relative_path: content} for env files.
ABSTRACT	`_generate_readme_scaffold(request, config, **kwargs)`	`str`	Return README.md skeleton as string.
PROTECTED	`_generate_license(request, config, **kwargs)`	`str|None`	Return LICENSE text. Default: Apache-2.0. None for PROPRIETARY.
ABSTRACT	`_generate_docker_config(request, config, **kwargs)`	`Dict[str,str]`	Return {'Dockerfile': content, '.dockerignore': content}.
PROTECTED	`_generate_makefile(request, config, **kwargs)`	`str|None`	Return Makefile content. Default: None. Override per runtime.
PROTECTED	`_validate_output(result, config, **kwargs)`	`bool`	Confirm required files created. Default: checks manifest + README.

ACCESS	METHOD SIGNATURE	RETURNS	PURPOSE / WHEN TO OVERRIDE
PROTECTED	`_handle_error(error, request, result, config, **kwargs)`	`ScaffoldResult`	Log error, append to errors[], return safe result.
PRIVATE	`__load_json_config(config_path)`	`ScaffoldRequest`	Read scaffold-config.json['scaffold']. Never override.
PRIVATE	`__load_prompt_config(prompt, config)`	`ScaffoldRequest`	LLM-based config parsing. Never override.
PRIVATE	`__write_file(root, relative_path, content)`	`str`	Write file + create parents. Returns absolute path.
PRIVATE	`__create_directory(path)`	`None`	mkdir -p. Idempotent.
PRIVATE	`__generate_scaffold_key(request)`	`str`	name:runtime:build_tool:platform — business idempotency key.
PRIVATE	`__is_already_scaffolded(key, config)`	`bool`	True if project dir exists and is non-empty.
PRIVATE	`__resolve_output_path(request, config)`	`str`	request > config > env BOOTSTRAP_OUTPUT_PATH > './'
PRIVATE	`__write_build_manifest(request, root, content)`	`str`	Write manifest to runtime-correct filename.
PRIVATE	`__build_dev_request(request, a2c_cfg)`	`dict`	Build A2C DevRequest for P0 -> A2C handoff.

4. INTERNAL EXECUTION SEQUENCE — BOOTSTRAP()

The sequence below is fixed and enforced by the base class. Subclasses never modify it. Every step is logged. A failure at any step calls _handle_error() and returns ScaffoldResult with success=False.

Step	Action	Calls	Condition
1	Resolve config from external source	_resolve_config()	Only if source = 'json' or 'prompt'
2	Validate ScaffoldRequest	_validate_request()	Always. Return False -> ValueError abort
3	Idempotency check	__is_already_scaffolded()	Only if self.idempotency = True
4	Build scaffold plan	_plan_scaffold()	Always. Result stored in request['scaffold_plan']
5	Create directory tree	_scaffold_structure()	Always
6	Generate + write build manifest	_generate_build_manifest() + __write_build_manifest()	Always
7	Generate dependency config	_generate_dependency_config()	Only if method returns non-None
8	Generate git config	_generate_git_config()	Only if request['git_init'] != False
9	Generate env templates	_generate_env_templates()	Always
10	Generate README	_generate_readme_scaffold()	Always
11	Generate LICENSE	_generate_license()	Only if license != 'PROPRIETARY'
12	Generate Docker config	_generate_docker_config()	Only if include_docker = True
13	Generate Makefile	_generate_makefile()	Only if include_makefile = True
14	Validate output	_validate_output()	Always. Return False -> ValueError abort
15	Latency SLO check	(internal)	Warn if > max_latency_ms (default 10,000 ms)
16	Structured observability log + A2C handoff	(internal)	Always. A2C handoff only if config['a2c']['enabled'] = True

5. NFR ENFORCEMENT — CORRECT BY STRUCTURE

P0 enforces Non-Functional Requirements structurally, not through prompts. Every generated artifact is correct by construction:

NFR	How P0 Enforces It	Artifact
Security — non-root container	Dockerfile uses multi-stage build + RUN useradd/adduser (UID 1001)	Dockerfile
Security — no secrets in git	_.gitignore excludes .env, *.key, chroma_db/, *.zip, .venv/	gitignore
Security — health probes	Dockerfile includes HEALTHCHECK pointing to /health or /actuator/health	Dockerfile
Observability — env-driven config	All LLM model IDs, cloud endpoints, OTel config in .env.example	env.example
Portability — multi-cloud	.env.example includes platform-specific vars for AWS/GCP/Azure	env.example
Clean Architecture — directory structure	src/{pkg}/models/, services/, routers/, tools/ matches E2A layers	Directory tree
CI/CD readiness	.github/workflows/ directory created; Makefile targets match pipeline stages	Makefile + .github/
Idempotency — same project not re-created	__is_already_scaffolded() checks project dir before any file writes	bootstrap() Step 3
Dev workflow — standard commands	Makefile: install, test, lint, build, run, docker-build, clean	Makefile
Dependency pinning	pyproject.toml/pom.xml includes all declared deps with version ranges	Build manifest
E2A attribution	README scaffold includes E2A framework link and Clean Architecture layer map	README.md

6. CONFIGURATION RESOLUTION — THREE INPUT SOURCES

ScaffoldRequest can be provided from three sources. All three produce the same ScaffoldRequest TypedDict. The resolution chain for any individual field follows the same priority as E2A:

Priority	Source	How to use
1 (highest)	Inline dict (source='inline')	Pass ScaffoldRequest directly to bootstrap(request, config)
2	JSON config file (source='json')	Set request['source']='json', request['config_path']='./scaffold-config.json'
3	Natural language prompt (source='prompt')	Set request['source']='prompt', request['prompt']='Create a FastAPI service on AWS...'

6.1 JSON Configuration File — scaffold-config.json

The JSON config file contains two top-level sections: **scaffold** (required, drives P0) and **a2c** (optional, drives A2C handoff via BootstrapAndGenerateWorkflow). Both can be present in the same file to run the full pipeline with one command.

```
# scaffold-config.json — Python/Poetry example (FastAPI on AWS)
{
  "scaffold": {
    "source":           "json",
    "runtime":          "python",
    "runtime_version": "3.12",
    "platform":         "aws",
    "build_tool":       "poetry",
    "project type":     "fastapi microservice",
    "project_name":     "SettlementEngine",
    "package_name":     "settlement_engine",
    "output_path":      "./generated/",
    "license":          "Apache-2.0",
    "include_docker":   true,
    "include_makefile":true,
    "dependencies": [
      "fastapi>=0.111.0", "uvicorn[standard]>=0.30.0", "pydantic>=2.7.0",
      "structlog>=24.1.0", "opentelemetry-sdk>=1.24.0", "boto3>=1.34.0",
      "langgraph>=0.1.0", "pybreaker>=1.0.1", "tenacity>=8.3.0"
    ],
    "dev_dependencies": ["pytest>=8.2.0","pytest-asyncio>=0.23.0","ruff>=0.4.0"]
  },
  "a2c": {
    "enabled":         true,
    "project_type":    "FastAPI Microservice",
    "mandatory_nfrs": ["Idempotency","Observability"],
    "target_cloud":    "aws",
    "context":         "Financial settlement microservice with saga orchestration"
  }
}
```

```
# scaffold-config.json — Java/Maven example (Spring Boot on AWS)
{
  "scaffold": {
    "source":                "json",
    "runtime":               "java",
    "runtime_version":       "21",
    "platform":              "aws",
    "build_tool":            "maven",
    "project_type":          "spring_boot",
    "project name":          "FinancialSettlementPlatform",
    "group_id":              "com.subhamviky",
    "artifact_id":           "financial-settlement-platform",
    "spring_boot_version":   "3.3.0",
    "java_source_version":   "21",
    "packaging":             "jar",
    "output_path":           "./generated/",
    "license":               "Apache-2.0",
    "include_docker":        true,
```

```
    "include_makefile":     true,
    "dependencies":         ["web","data-jpa","kafka","actuator","data-redis"]
  },
  "a2c": {
    "enabled":        true,
    "project_type":   "Spring Boot",
    "mandatory_nfrs": ["Idempotency","Observability","CircuitBreaker"],
    "target_cloud":   "aws",
    "context":        "Java settlement platform: Saga, @Idempotent AOP, double-entry ledger"
  }
}
```

7. GENERATED DIRECTORY STRUCTURES

7.1 Python FastAPI Microservice (Poetry)

```
SettlementEngine/
  pyproject.toml          Poetry build manifest with all dependencies
  .gitignore              Python-specific: __pycache__, .venv, .env, chroma_db, *.zip
  .env.example            AWS/GCP/Azure env vars + E2A NFR config
  README.md               E2A spike framing + stack badges + structure tree
  LICENSE                 Apache-2.0
  Dockerfile              Multi-stage Python build, non-root UID 1001, HEALTHCHECK
  .dockerignore           .venv, __pycache__, .env, .git, tests
  Makefile                install, test, lint, build, run, docker-build, clean
  .github/
    workflows/            CI/CD workflow directory (A2C fills this in Phase 4)
  src/
    settlement_engine/
      __init__.py
      main.py             FastAPI app entry point placeholder
      config/             Configuration classes
        __init__.py
      models/             Pydantic BaseModel domain entities (E2A: AgentState equivalent)
        __init__.py
      services/           Business logic classes (E2A: BIMP / @Service equivalent)
        __init__.py
      routers/            FastAPI APIRouter (E2A: OData Service Binding equivalent)
        __init__.py
      tools/              MCP tool classes (E2A: BaseToolService subclasses)
        __init__.py
  tests/
    __init__.py
    conftest.py           Pytest fixtures
```

7.2 Python LangGraph Agent (Poetry)

Same base structure as FastAPI microservice plus:

```
SettlementEngine/
  ...  (all files above)
  src/settlement_engine/
    agents/               BaseAgent subclasses (RouterAgent, KnowledgeAgent, etc.)
      __init__.py
    rag/                  BaseRAGPipeline subclasses
      __init__.py
    workflows/            BaseWorkflow subclasses (LangGraph StateGraph)
        init  .py
```

7.3 Java Spring Boot (Maven)

```
FinancialSettlementPlatform/
  pom.xml                 Maven POM with Spring Boot parent, all dependencies
  .gitignore              Java: target/, *.class, .idea/, *.jar, .gradle/
  README.md               E2A spike framing + Spring Boot stack badges
  LICENSE                 Apache-2.0
  Dockerfile              Multi-stage eclipse-temurin build, non-root, HEALTHCHECK
  .dockerignore           target/, .git, .idea, *.md
  Makefile                build, test, run, docker-build, clean
  .github/
    workflows/            CI/CD directory (A2C fills in Phase 4)
  src/
    main/
      java/com/subhamviky/
        config/           Spring @Configuration classes
        domain/           JPA @Entity classes (E2A: CDS View Entity equivalent)
        service/          @Service classes (E2A: BIMP equivalent)
        controller/       @RestController (E2A: OData Service Binding equivalent)
        repository/       Spring Data JPA repositories
      resources/
```

```
      application.yml         Default Spring Boot config with actuator, logging
      application-dev.yml     Dev profile (show-sql: true)
      application-prod.yml    Prod profile (logging WARN)
  test/
    java/com/subhamviky/
      FinancialSettlementPlatformApplicationTests.java
```

8. CONCRETE IMPLEMENTATIONS — QUICK REFERENCE

Class	Runtime	Build Tool	Project Types Supported	Key Overrides
PythonPoetryBootstrapper	Python 3.x	Poetry	fastapi_microservice, lambda, langgraph_agent, library	All abstract methods. _generate_makefile(): poetry commands.
PythonPipBootstrapper	Python 3.x	pip	fastapi_microservice, lambda, library	_generate_build_manifest(): setup.cfg. _generate_dependency_config(): requirements.txt.
JavaMavenBootstrapper	Java 21+	Maven	spring_boot, library	All abstract methods. pom.xml with spring-boot-starter-parent.
JavaGradleBootstrapper	Java 21+	Gradle	spring_boot, library	_generate_build_manifest(): build.gradle.kts. _generate_makefile(): gradle wrapper commands.
NodeNpmBootstrapper	Node.js 20+	npm	express_api, next_app, library	_generate_build_manifest(): package.json with scripts.
GoModulesBootstrapper	Go 1.22+	go_modules	http_service, cli, library	_generate_build_manifest(): go.mod. _scaffold_structure(): cmd/, internal/, pkg/ layout.

8.1 Factory — ProjectBootstrapperFactory

The factory resolves the correct subclass from a ScaffoldRequest without the caller knowing which class to instantiate. It follows the same approved-list governance pattern as E2A agent routing.

```
# Resolve and run — single call
bootstrapper = ProjectBootstrapperFactory.create(request, config)
result = bootstrapper.bootstrap(request, config)

# Register a new bootstrapper (call at module import time)
ProjectBootstrapperFactory.register('go', 'go_modules', GoModulesBootstrapper)
```

9. BOOTSTRAPANDGENERATEWORKFLOW — COMPOSING P0 WITH A2C

BootstrapAndGenerateWorkflow orchestrates the complete pipeline: P0 scaffold followed by A2C code generation. It reads a single scaffold-config.json that contains both the scaffold section (for P0) and the a2c section (for A2C). Three operating modes:

Mode	How to invoke	What runs
Bootstrap only	execute(request, config, bootstrap_only=True)	P0 only — creates project structure, exits
Generate only	execute(request, config, generate_only=True)	A2C only — assumes project already bootstrapped
Full pipeline (default)	execute(request, config)	P0 scaffold -> A2C code generation in sequence

9.1 Full Pipeline Execution Sequence

```
BootstrapAndGenerateWorkflow.execute(scaffold_request, config)

  Step A — P0 Phase (BaseProjectBootstrapper.bootstrap)
    1  resolve config (JSON/prompt/inline)
    2  validate ScaffoldRequest
    3  idempotency check
    4  create directory structure
    5  generate build manifest (pom.xml / pyproject.toml)
    6  generate git config, env templates, README, LICENSE, Dockerfile, Makefile
    7  validate output
    8  build dev request for A2C handoff

  Step B — A2C Phase (SDLCAssistantAgent.run)  — only if a2c.enabled=true
    1  RequirementsAgent  validates project_type, mandatory_nfrs
    2  IaCAgent           generates Terraform (ECS/EKS/Cloud Run)
    3  CodeGenAgent       generates Clean Architecture microservice code
    4  CICDAgent          generates GitHub Actions with OIDC, RAGAS, Trivy
    5  CodeCriticAgent    validates NFR completeness score >= 0.75

  Output: {scaffold_result: ScaffoldResult, generation_result: DevRequest}
```

9.2 A2C Handoff — dev_request Population

When config['a2c']['enabled'] is True, bootstrap()'s Step 16 (observability log) calls the private __build_dev_request() method, which maps ScaffoldRequest fields to A2C's DevRequest TypedDict. Fields from both the scaffold section and the a2c section contribute to the handoff payload:

DevRequest field	Source
project_name	From ScaffoldRequest['project_name']
language	From ScaffoldRequest['runtime'] (python/java)
target_cloud	From ScaffoldRequest['platform'] (aws/gcp/azure)
project_type	From config['a2c']['project_type']
mandatory_nfrs	From config['a2c']['mandatory_nfrs'] (A2C adds missing mandatory ones)
context	From config['a2c']['context']

10. USAGE PATTERNS

Pattern 1 — Bootstrap only (inline request)

```
from base_project_bootstrapper import ProjectBootstrapperFactory

request = {
    'source':          'inline',
    'runtime':         'python',
    'runtime_version': '3.12',
    'platform':        'aws',
    'build_tool':      'poetry',
    'project_type':    'fastapi_microservice',
    'project_name':    'SettlementEngine',
    'package_name':    'settlement_engine',
    'license':         'Apache-2.0',
    'include_docker':  True,
    'dependencies':    ['fastapi>=0.111', 'pydantic>=2.7', 'structlog>=24.1'],
    'dev_dependencies':['pytest>=8', 'ruff>=0.4'],
}

bootstrapper = ProjectBootstrapperFactory.create(request)
result = bootstrapper.bootstrap(request)

print(result['project_root'])         # /abs/path/SettlementEngine
print(len(result['scaffolded_files'])) # e.g. 18
print(result['success'])              # True
```

Pattern 2 — Bootstrap from JSON file

```
request = { 'source': 'json', 'config_path': './scaffold-config.json' }
bootstrapper = ProjectBootstrapperFactory.create(
    {'runtime': 'python', 'build_tool': 'poetry'}  # factory needs these to route
)
result = bootstrapper.bootstrap(request)
```

Pattern 3 — Full pipeline: P0 scaffold + A2C generation

```
from base_project_bootstrapper import BootstrapAndGenerateWorkflow

config = {
    'output_path': './generated/',
    'a2c': {
        'enabled':        True,
        'project_type':   'FastAPI Microservice',
        'mandatory_nfrs': ['Idempotency', 'Observability'],
        'target_cloud':   'aws',
        'context':        'Financial settlement with saga orchestration'
    }
}

request = {
    'source': 'json',
    'config_path': './scaffold-config.json',
    'runtime': 'python',
    'build_tool': 'poetry',
}

workflow = BootstrapAndGenerateWorkflow(config=config)
result = workflow.execute(request, config)

print(result['scaffold_result']['scaffolded_files'])
print(result['generation_result']['dev_request'])
```

Pattern 4 — Disable idempotency (re-scaffold existing project)

```
bootstrapper = PythonPoetryBootstrapper(config=config)
bootstrapper.idempotency = False   # override: re-scaffold even if dir exists
result = bootstrapper.bootstrap(request, config)
```

Pattern 5 — Prompt-driven scaffolding (LLM config resolution)

```
request = {
    'source': 'prompt',
    'prompt': 'Create a Python FastAPI microservice on AWS for payment reconciliation.'
              ' Use Poetry. Include Docker and Makefile. License: Apache-2.0.',
    'runtime':     'python',   # factory routing still needs these
    'build_tool':  'poetry',
}
config = { 'model id': 'anthropic.claude-3-5-sonnet-20241022-v2:0' }

bootstrapper = PythonPoetryBootstrapper(config=config)
result = bootstrapper.bootstrap(request, config)
```

11. IMPLEMENTATION CHECKLIST — NEW BOOTSTRAPPER SUBCLASS

Follow this checklist when adding a new runtime or build tool. All items are mandatory except those marked optional.

#	Task	Mandatory?
1	Define bootstrapper_name class attribute (e.g. 'NodeNpmBootstrapper')	Yes
2	Call ProjectBootstrapperFactory.register(runtime, build_tool, MyBootstrapper)	Yes
3	Override _validate_request(): check required fields for this runtime	Yes
4	Override _resolve_config(): handle source='json' + source='prompt' loading	Yes
5	Override _plan_scaffold(): return [{step, file, description}] task list	Yes
6	Override _scaffold_structure(): call __create_directory() for every dir in the tree	Yes
7	Override _generate_build_manifest(): return full file content as string	Yes
8	Override _generate_git_config(): runtime-correct .gitignore (exclude build artifacts, .env, IDE)	Yes
9	Override _generate_env_templates(): {relative_path: content} for all env config files	Yes
10	Override _generate_readme_scaffold(): project name, badges, E2A link, structure tree, get-started	Yes
11	Override _generate_docker_config(): multi-stage, non-root UID 1001, HEALTHCHECK	Yes
12	Override _generate_dependency_config() + _dependency_file_name() if deps not in manifest	If pip/yarn
13	Override _generate_makefile(): standard targets (install, test, lint, build, run, docker-build, clean)	Recommended
14	Override _validate_output(): add runtime-specific checks beyond manifest + README	Recommended
15	Override _generate_license() only to add full license text; default Apache-2.0 is provided	Optional
16	Override _handle_error() to add alerting, cleanup, or partial-state recovery	Recommended
17	DO NOT override bootstrap() — the 16-step sequence is enforced by the base class	Never
18	DO NOT call any private method (__write_file etc.) from outside the class	Never

12. ENVIRONMENT VARIABLE REFERENCE

Variable	Used in	Default	Controls
BOOTSTRAP_OUTPUT_PATH	__resolve_output_path()	'./'	Base directory for all generated projects
BOOTSTRAP_MAX_LATENCY_MS	bootstrap() Step 15	10000	Latency SLO threshold in milliseconds
LLM_MODEL_ID	__load_prompt_config()	'default'	LLM model for prompt-based config resolution
TOOL_TIMEOUT	BaseToolService (A2C)	5.0	HTTP timeout for A2C tool calls (inherited)
MIN_CONFIDENCE	BaseAgent (A2C)	0.85	RAGAS groundedness gate (inherited from E2A)

13. P0 IN THE FULL E2A / A2C / P0 FRAMEWORK CONTEXT

The three frameworks form a layered stack. Each addresses a distinct concern. All follow the same E2A Template Method Pattern.

Framework	Concern	Entry point	Input	Output
E2A	Build governed agentic AI systems	agent.run(state, config)	AgentState	Updated AgentState with response, tokens, trace_id
A2C	Generate governed microservice code using E2A agents	sdlc_agent.run(dev_request, config)	DevRequest	generated_code, generated_iac, generated_cicd, critic_report
P0 (this document)	Generate project structure before A2C runs	bootstrapper.bootstrap(request, config)	ScaffoldRequest	project_root, pom.xml/pyproject.toml, Dockerfile, Makefile, README

```
Developer has an idea for a new microservice
            |
            v
  [P0]  ProjectBootstrapperFactory.create(request)
        bootstrapper.bootstrap(request, config)
            |
            |  generates: pyproject.toml / pom.xml, directory tree,
            |             Dockerfile, Makefile, .gitignore, .env.example
            |
            v
  [A2C]  SDLCWorkflow.execute(dev_request, config)
         RequirementsAgent -> CodeGenAgent -> IaCAgent -> CICDAgent -> CodeCriticAgent
            |
            |  generates: service classes, RAG pipeline, Terraform, GitHub Actions
            |
            v
  [E2A]  All generated agent/service code inherits BaseAgent, BaseRAGPipeline,
          BaseToolService — NFRs enforced as structural contracts at runtime
```